

The $\theta \in [\frac{\pi}{2}, \pi]$ case can also be readily obtained, and the conditions of (5) are hence satisfied for all the possible cases. We can also apprehend that there are two orthogonal pairs $(q, \omega q)$ and $(\omega^2 q, \omega^3 q)$, thus counting the four dimensions of the space, which are not equivalent because their imaginary unit is different. Adopting these ideas, we generally have

$$q(\theta_1 + \theta_2) = \begin{cases} \cos \theta_0 + \omega \sin \theta_0 & n \text{ even} & \theta_0 \in [0, \pi] \\ \cos(\pi - \theta_0) + \tilde{\omega} \sin(\pi - \theta_0) & n \text{ odd,} & \theta_0 \in [0, \frac{\pi}{2}] \\ \cos(\frac{\pi}{2} - \theta_0) + \tilde{\omega} \sin(\frac{\pi}{2} - \theta_0) & n \text{ odd,} & \theta_0 \in [\frac{\pi}{2}, \pi] \\ \theta_1 + \theta_2 = n\pi + \theta_0 & \text{and} & \tilde{\omega} = -\omega \end{cases} \quad (16)$$

Replicating this procedure using the symplectic notation (8), we obtain

$$\left. \begin{aligned} q &= \cos \vartheta e^{i\phi} + \sin \vartheta e^{i\psi} j \\ qj &= \cos(\frac{\pi}{2} - \vartheta) e^{i(\psi-\pi)} + \sin(\frac{\pi}{2} - \vartheta) e^{i\phi} j \\ qj^2 &= \cos \vartheta e^{i(\phi-\pi)} + \sin \vartheta e^{i(\psi-\pi)} j \\ qj^3 &= \cos \vartheta (\frac{\pi}{2} - \vartheta) e^{i\psi} + \sin(\frac{\pi}{2} - \vartheta) e^{i(\phi-\pi)} j \end{aligned} \right\} \quad \vartheta \in [0, \frac{\pi}{2}] \quad (17)$$

and also

$$q(\vartheta_1 + \vartheta_2) = \begin{cases} \cos \vartheta_0 e^{i\phi} + \sin \vartheta_0 e^{i\psi} j & n = 0 \bmod 4 \\ \cos(\frac{\pi}{2} - \vartheta_0) e^{i(\psi-\pi)} + \sin(\frac{\pi}{2} - \vartheta_0) e^{i\phi} j & n = 1 \bmod 4 \\ \cos \vartheta_0 e^{i(\psi-\pi)} + \sin \vartheta_0 e^{i(\phi-\pi)} j & n = 2 \bmod 4 \\ \cos \vartheta (\frac{\pi}{2} - \vartheta_0) e^{i\psi} + \sin(\frac{\pi}{2} - \vartheta_0) e^{i(\phi-\pi)} j & n = 3 \bmod 4 \end{cases} \quad (18)$$

where

$$\vartheta_1 + \vartheta_2 = \vartheta_0 + n\frac{\pi}{2}, \quad \vartheta_0 \in [0, \frac{\pi}{2}] \quad \text{and} \quad n \in \mathbb{N}.$$

Therefore the set of necessary quaternionic concepts is complete, comprising the polar notations, the inner product, and the requirements for orthogonality.

3 ANGULAR FUNCTION AND WINDING NUMBER

We separate the analysis according to the quaternionic notations defined in the Section 2: the polar cartesian notation (4-5) and the symplectic notation (7-8) for quaternion curves, and therefore the global property of winding number that is defined to plane curves in $\mathbb{R}^2$ [22] will be extended to curves parametrized in quaternionic terms.

3.1 Polar cartesian quaternionic curves

The parametrization of quaternionic curves in the $\mathbb{H} \equiv \mathbb{R} \times \mathbb{R}\omega$ space, where (4-5) hold, is similar to the complex description of plane curves in the $\mathbb{R} \times \mathbb{R}i$ plane, but the quaternionic case can be considered as a half space, and the imaginary unit will be generalized to a function $\omega : I \rightarrow \mathbb{H}$, where we define to the rest of the paper the real interval

$$I = [a, b], \quad \text{and their variable} \quad t \in I. \quad (19)$$

The concept of quaternionic curve is analogous to regular real plane curves either in $\mathbb{R}^2$ or in $\mathbb{C}$, so that

Definition 3.1 (Regular quaternionic curves) *The parametrized curve $q : I \rightarrow \mathbb{H}$ is regular in $I \subset \mathbb{R}$ if $|q'(t)| \neq 0 \quad \forall t \in I$.*

Therefore, we can prove that

Proposition 3.1 (Imaginary quaternionic function) *Let $x = x(t)$ be a pure imaginary regular quaternionic curve. Therefore,*

- i. $\overline{\omega}' = -\omega'$
- ii. $\omega\omega' = -\omega'\omega \quad \text{where} \quad \omega = \frac{x}{|x|}$
- iii. $(\omega')^2 = -\frac{1}{|x|^2}(|x'| - |x'|)^2$
- iv. $\omega\omega'' + \omega''\omega = 2|\omega'|^2$

Proof: The item (ii) comes immediately from the derivative of $\omega^2 = -1$, and (ii) comes from

$$\omega' = \frac{|x'|}{|x|}\omega - \frac{|x|'}{|x|}\omega, \quad \text{where} \quad \omega = \frac{x}{|x|}.$$

where $\overline{\omega} = -\omega$. Item (iii) comes after squaring (iii) and using identity

$$\omega\omega + \omega\omega = -2\frac{|x'|}{|x'|}.$$

Property (iv) is also immediate after deriving (i) and using (ii). □

Proposition (3.1) establishes the relevant properties of the quaternionic imaginary unit in polar notation (4). Nonetheless, the definition of an angular function is also needed. In the case of a plane curve, an arbitrary point chosen out of their trace will be the vertex for the measured angle between two points of the curve. Keeping this previous concept, a suitable expression for the angular function is as follows:

Proposition 3.2 (The polar angular function) *Let $q : I \rightarrow \mathbb{H}$ be a regular quaternionic curve. The polar angular function of $q(t)$ around the regular curve $p_0(t)$ out of the trace of the curve q is*

$$\theta(t) = \int_a^t \frac{\langle p(\tau)\omega(\tau), p'(\tau) \rangle}{|p(\tau)|^2} d\tau \quad \text{where} \quad p(\tau) = q(\tau) - p_0(\tau), \quad (20)$$

and $\omega(t)$ is the imaginary unit of the polar parametrization of $p(t)$.

Proof: The parametrization

$$p(t) - p_0(t) = \rho(t) \left(\cos \theta(t) + \sin \theta(t) \omega(t) \right),$$

leads to

$$\langle (p - p_0)\omega, (p - p_0)' \rangle = \rho^2 \theta'$$

whose integration immediately gives the desired result. □

Interestingly, the imaginary unit ω does not affect the result. By way of example, let us consider the quaternion curve

$$q(t) = R\omega(t) + R[\cos t + \omega(t)\sin t], \quad t \in [0, 2n\pi], \quad n \in \mathbb{Z} \setminus \{0\}. \quad (21)$$

This curve can be seen as a sort of spiral whose projection over the $\mathbb{R} \times \mathbb{R}\omega$ plane is identical to a circumference of radius R and center $R\omega$ for all values of the parameter t . Hence

$$p(t) = q(t) - p_0(t) \quad \Rightarrow \quad \langle p\omega, p' \rangle = R^2. \quad (22)$$

To obtain the polar angle swept along the interval, and to keep the positive character expected to the pure imaginary axis in the polar notation (5), two separate integrals are needed. A first one for $t \in [0, \pi]$, where the imaginary direction is ω , and a second integration for $t \in [\pi, 2\pi]$. This second integral goes to the correct parametrization because $[\pi, 2\pi] \rightarrow [\pi, 0]$ and $\omega \rightarrow -\omega$. Consequently $\theta = 2n\pi$, after n integrations, a perfect agreement to the complex plane after the transformation $\omega \rightarrow i$. After this example, we define

Definition 3.2 (The winding number) Let $q : I \rightarrow \mathbb{H}$ be a continuous closed and piecewise differentiable quaternionic curve. The winding number of this curve along the continuous and differentiable $p_0(t) : I \rightarrow \mathbb{H}$ is

$$\mathcal{R}(q, p_0) = \frac{1}{2\pi} \int_a^b \frac{\langle p(\tau)\omega, p'(\tau) \rangle}{|p(\tau)|^2} d\tau$$

where

$$p(t) = q(t) - p_0(t), \quad p(t) \in \mathbb{R} \times \mathbb{R}\omega, \quad \text{and} \quad \mathcal{R}(q, p_0) \in \mathbb{Z}.$$

□

The analogy between complex curves and quaternionic curves is thus completed. In the complex case a fixed point is chosen, while in the quaternionic case a continuous and smooth quaternionic curve is the reference to the winding number. We observe that a variable radius of the curve does not change the result, as was seen in Proposition 3.2. Moreover, we can prove that:

Proposition 3.3 Let $q : I \rightarrow \mathbb{H}$ be a closed regular curve. Therefore, $\exists \rho > 0$ such that, if $p(t) : I \rightarrow \mathbb{H}$ is a regular curve that satisfies $|p(t)| \geq \rho$, then

$$\mathcal{R}(p, q) = 0.$$

Proof: Let ρ_0 be the maximum value of $|q(t)|$, and $|p(\tau)| = \rho$ the minimum value of $|p(t)|$. If $\rho > \rho_0$, $p(\tau)\omega(\tau) \cap q(t) = \emptyset$, and the polar angular function $\theta(t)$ defined by $q(t) - p(t)$ is such that $\theta(t) < \pi/2$, and hence $\theta(b) = 0$.

□

3.2 Symplectic quaternionic curves

The parametrization of quaternionic curves in the $\mathbb{C} \times \mathbb{C}j$ plane is different to the description of quaternionic curves parametrized in $\mathbb{R} \times \mathbb{R}\omega$ plane. If we naively try to repeat the proof of Proposition (3.2), defining

$$p(t) = \rho(\cos \vartheta e^{i\phi} + \sin \vartheta e^{i\psi} j), \quad \text{where} \quad t \in I = [a, b] \quad (23)$$

and $\vartheta(t)$, $\phi(t)$ and $\psi(t)$ are real functions, we get

$$\langle pj, p' \rangle = \rho^2 [\vartheta' \cos(\phi - \psi) + \sin \vartheta \cos \vartheta \sin(\phi - \psi)(\phi + \psi)'], \quad (24)$$